

Ehsan Sharafian

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PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

EDUCATION

Ph.D. in Mechanical Engineering University of Maine, Orono, ME	Expected Dec 2027
GPA: 4.00/4.00 Dissertation: Smart Real-Time Gait Training System	
M.S. in Mechanical Engineering Amirkabir University of Technology (Tehran Polytechnic)	Feb 2020
GPA: 3.65/4.00 Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces	
B.S. in Mechanical Engineering University of Tehran	Jul 2017

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS
Mobile & Software	MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules
Biomechanics & Signal Processing	Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim
Robotics	Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant Biorobotics and Biomechanics Lab, University of Maine Orono, ME	Sep 2023 – Present
- Led the development of real-time wearable sensing systems and Android-based mobile applications for gait analysis, activity recognition, fall-risk assessment, and home-based rehabilitation. - Built a smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify locomotion across level walking, stairs, and ramps, achieving 96.08% accuracy, <1.1 cm error, and 112 ms latency. - Engineered a lightweight human activity recognition system using two IMUs, biomechanical feature extraction, and classifiers including SVM and NB, achieving 99.1% accuracy for normal locomotion and 93.1% accuracy for atypical stair-negotiation strategies in older adults. - Architected a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry for adults aged 65+. - Demonstrated up to 30% improvement in stride length and walking speed during a single-session haptic feedback study, with minimal cognitive demand. - Led IRB-approved human-subject validation studies with 85+ participants across gait analysis, activity recognition, and haptic feedback protocols. - Developed real-time algorithms and data pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.	

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Coordinated hands-on validation sessions with **6 experienced surgeons**, using training, troubleshooting, and usability feedback to refine surgeon-facing robotic system workflows.

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the 9th International Conference on Robotics and Mechatronics, including lab demonstrations, equipment readiness, and presentation logistics.

SELECTED HONORS

Flagship Fellowship Award, University of Maine	Sep 2024 – Sep 2025
Graduate Student Government Award for conference proposal	Mar 2025

LEADERSHIP & SERVICE

Graduate Senator, Department of Mechanical Engineering, University of Maine	Sep 2025 – Present
President, Iranian Graduate Student Association, University of Maine	Sep 2024 – Apr 2026

PUBLICATIONS

- 6 peer-reviewed (5 first-author) journal articles in wearable sensing, biomechanics, rehabilitation engineering, and robotics.
- **Sensors** (2026) – Foot-height estimation and locomotion classification using a foot-mounted IMU and smartphone.
 - **IEEE Access** (2025) – IMU-based activity recognition for older adults using minimal biomechanical features.
 - **IEEE TNSRE** (2025) – Real-time haptic feedback effects on gait performance and cognitive load.
 - **Journal of Biomechanics** (2025) – Thigh extension and leg coordination in young and older adult walkers.
 - **International Journal of Robotics and Automation** (2023) – Robot-arm trajectory optimization for production-line applications.
 - **Robotica** (2022) – Screw theory-based constraint wrench analysis for a Delta manipulator.